

FX App API

API Go pour la gestion de transactions de change, avec profils KYC clients, analyse AML/KYC, scoring de risque et alertes 'risk_flags'.

Fonctionnalites

- Gestion des cambistes/traders.
- Gestion des clients.
- Creation de client avec profil KYC initial.
- Consultation et mise a jour du profil KYC.
- Recalcul manuel du risque KYC client.
- Creation de transaction FX avec conversion calculee depuis le taux fixe par le trader.
- Analyse AML/KYC synchrone apres creation de transaction.
- Creation automatique de 'risk_flags' quand une regle AML/KYC est declenchee.
- Consultation, filtrage, creation manuelle et suppression logique des 'risk_flags'.
- Initialisation automatique des tables PostgreSQL au demarrage.

Stack

- Go '1.26.2'
- PostgreSQL
- Driver SQL 'github.com/lib/pq'
- Serveur HTTP natif 'net/http'
- Port par default '8080'

Demarrage

```
'''bash
make run
'''
```

Equivalent:

```
'''bash
go build -o bin/fx-app ./cmd
./bin/fx-app
'''
```

Au demarrage, l'application:

1. charge '.env' si le fichier existe;
2. ouvre la connexion PostgreSQL;
3. cree la base cible si elle n'existe pas;
4. initialise les tables;
5. expose l'API sur 'http://localhost:8080'.

Configuration

'DATABASE_URL' est prioritaire si elle est definie. Sinon l'application compose une chaine de connexion avec les variables suivantes.

Variable	Defaut	Description
'DATABASE_URL'	vide	URL PostgreSQL complete.
'DB_HOST'	'localhost'	Hote PostgreSQL.
'DB_PORT'	'5432'	Port PostgreSQL.
'DB_USER'	utilisateur systeme courant	Utilisateur PostgreSQL.

'DB_PASSWORD'	vide	Mot de passe PostgreSQL.
'DB_NAME'	'forex_trader_kyc_db'	Nom de la base applicative.
'DB_SSLMODE'	'disable'	Mode SSL PostgreSQL.

Exemple '.env':

```
'''env
DB_HOST=localhost
DB_PORT=5432
DB_USER=postgres
DB_PASSWORD=postgres
DB_NAME=forex_trader_kyc_db
DB_SSLMODE=disable
'''
```

Format Des Erreurs

Les erreurs HTTP sont renvoyees en JSON:

```
'''json
{
  "message": "description de l'erreur"
}
'''
```

Statuts utilises:

Statut	Cas
---	---
'400 Bad Request'	JSON invalide, id invalide, champ obligatoire absent.
'404 Not Found'	Ressource introuvable sur les routes de lecture/suppression.
'500 Internal Server Error'	Erreur repository, contrainte SQL, erreur de persistance ou erreur metier non mappee.

Authentification

Le helper 'MakeProtectedHttpHandleFunc' existe et attend:

```
'''http
Authorization: Bearer <token>
'''
```

Aucune route actuelle ne l'utilise encore. La verification réelle du token est commentee.

Routes

Methode	Route	Description
---	---	---
'POST'	'/traders'	Creer un cambiste.
'GET'	'/traders/{id}'	Recuperer un cambiste.
'PUT'	'/traders/{id}'	Mettre a jour un cambiste.
'DELETE'	'/traders/{id}'	Supprimer logiquement un cambiste.
'POST'	'/customers'	Creer un client simple.
'POST'	'/customers/kyc'	Creer un client avec profil KYC.
'GET'	'/customers/{id}'	Recuperer un client.
'PUT'	'/customers/{id}'	Mettre a jour un client.
'GET'	'/customers/{id}/kyc'	Recuperer le profil KYC d'un client.
'PUT'	'/customers/{id}/kyc'	Mettre a jour le KYC et recalculer le risque.

```
| 'POST' | '/customers/{id}/kyc/reassess' | Recalculer le risque KYC du client. |
| 'POST' | '/transactions' | Creer une transaction avec conversion. |
| 'GET' | '/transactions/{id}' | Recuperer une transaction. |
| 'POST' | '/risk-flags' | Creer une alerte de risque manuelle. |
| 'GET' | '/risk-flags' | Lister les alertes de risque. |
| 'GET' | '/risk-flags/{id}' | Recuperer une alerte de risque. |
| 'DELETE' | '/risk-flags/{id}' | Supprimer logiquement une alerte de risque. |
```

Catalogue Des Requetes JSON

Cette section resume toutes les routes avec leurs payloads. Les routes 'GET' et 'DELETE' n'ont pas de body JSON.

'POST /traders'

Body:

```
'''json
{
  "name": "Bureau Gombe",
  "first_name": "Jean",
  "last_name": "Kabila",
  "email": "jean@example.com",
  "phone": "+2438100000000",
  "status": "ACTIVE",
  "role": "TRADER",
  "password_hash": "$2a$...",
  "store_id": 1,
  "is_active": true
}
'''
```

Response '201 Created':

```
'''json
{
  "id": 1,
  "name": "Bureau Gombe",
  "first_name": "Jean",
  "last_name": "Kabila",
  "email": "jean@example.com",
  "phone": "+2438100000000",
  "status": "ACTIVE",
  "created_at": "2026-08-21T12:30:00Z",
  "updated_at": "2026-08-21T12:30:00Z",
  "deleted_at": null,
  "role": "TRADER",
  "password_hash": "$2a$...",
  "store_id": 1,
  "is_active": true
}
'''
```

'GET /traders/{id}'

Body: aucun.

Response '200 OK': objet 'Trader'.

```
'''json
{
  "id": 1,
  "name": "Bureau Gombe",
  "first_name": "Jean",
  "last_name": "Kabila",
  "email": "jean@example.com",
  "phone": "+2438100000000",
  "status": "ACTIVE",
  "created_at": "2026-08-21T12:30:00Z",
  "updated_at": "2026-08-21T12:30:00Z",
  "deleted_at": null,
  "role": "TRADER",
  "password_hash": "$2a$...",
  "store_id": 1,
  "is_active": true
}
'''
```

'PUT /traders/{id}'

Body:

```
'''json
{
  "name": "Bureau Gombe",
  "first_name": "Jean",
  "last_name": "Kabila",
  "email": "jean@example.com",
  "phone": "+2438100000000",
  "status": "ACTIVE",
  "role": "MANAGER",
  "password_hash": "",
  "store_id": 1,
  "is_active": true
}
'''
```

Response '200 OK': objet 'Trader'.

'DELETE /traders/{id}'

Body: aucun.

Response: '204 No Content'.

'POST /customers'

Body:

```
'''json
{
  "full_name": "Grace Hopper",
  "id_number": "ID-123456",
  "id_type": "PASSPORT",
  "phone": "+2438200000000",
  "address": "Kinshasa"
```

```
}  
'''
```

Response '201 Created':

```
'''json  
{  
  "id": 1,  
  "full_name": "Grace Hopper",  
  "id_number": "ID-123456",  
  "id_type": "PASSPORT",  
  "phone": "+2438200000000",  
  "address": "Kinshasa",  
  "created_at": "2026-08-21T12:30:00Z",  
  "updated_at": "2026-08-21T12:30:00Z",  
  "risk_level": "",  
  "risk_score": 0  
}  
'''
```

'POST /customers/kyc'

Body:

```
'''json  
{  
  "customer": {  
    "full_name": "Grace Hopper",  
    "id_number": "ID-123456",  
    "id_type": "PASSPORT",  
    "phone": "+2438200000000",  
    "address": "Kinshasa"  
  },  
  "kyc_profile": {  
    "legal_nature": "INDIVIDUAL",  
    "activity_profile": "BUSINESS",  
    "profession": "Entrepreneur",  
    "employer": "",  
    "company_name": "",  
    "registration_number": "",  
    "source_of_funds": "Business revenue",  
    "purpose_of_operations": "Currency exchange",  
    "expected_volume": 50000,  
    "expected_frequency": 12,  
    "last_verification_date": "2026-08-21T12:30:00Z"  
  }  
}  
'''
```

Response '201 Created': objet 'Customer'.

```
'''json  
{  
  "id": 1,  
  "full_name": "Grace Hopper",  
  "id_number": "ID-123456",  
  "id_type": "PASSPORT",  
  "phone": "+2438200000000",
```

```
"address": "Kinshasa",
"created_at": "2026-08-21T12:30:00Z",
"updated_at": "2026-08-21T12:30:00Z",
"risk_level": "LOW",
"risk_score": 20
}
'''
```

'GET /customers/{id}'

Body: aucun.

Response '200 OK': objet 'Customer'.

```
'''json
{
  "id": 1,
  "full_name": "Grace Hopper",
  "id_number": "ID-123456",
  "id_type": "PASSPORT",
  "phone": "+2438200000000",
  "address": "Kinshasa",
  "created_at": "2026-08-21T12:30:00Z",
  "updated_at": "2026-08-21T12:30:00Z",
  "deleted_at": null,
  "risk_level": "LOW",
  "risk_score": 20
}
'''
```

'PUT /customers/{id}'

Body:

```
'''json
{
  "full_name": "Grace Hopper",
  "id_number": "ID-123456",
  "id_type": "NATIONAL_ID",
  "phone": "+2438200000000",
  "address": "Gombe, Kinshasa"
}
'''
```

Response '200 OK': objet 'Customer'.

'GET /customers/{id}/kyc'

Body: aucun.

Response '200 OK': objet 'CustomerKYCProfile'.

```
'''json
{
  "id": 1,
  "customer_id": 1,
  "legal_nature": "INDIVIDUAL",
  "activity_profile": "BUSINESS",

```

```
"kyc_status": "UNVERIFIED",
"profession": "Entrepreneur",
"employer": "",
"company_name": "",
"registration_number": "",
"source_of_funds": "Business revenue",
"purpose_of_operations": "Currency exchange",
"expected_volume": 50000,
"expected_frequency": 12,
"last_verification_date": "2026-08-21T12:30:00Z",
"created_at": "2026-08-21T12:30:00Z",
"updated_at": "2026-08-21T12:30:00Z"
}
,,,
```

'PUT /customers/{id}/kyc'

Body:

```
'''json
{
  "legal_nature": "INDIVIDUAL",
  "activity_profile": "PROFESSIONAL",
  "kyc_status": "VERIFIED",
  "profession": "Consultant",
  "employer": "Self-employed",
  "company_name": "",
  "registration_number": "",
  "source_of_funds": "Consulting income",
  "purpose_of_operations": "FX operations",
  "expected_volume": 25000,
  "expected_frequency": 8,
  "last_verification_date": "2026-08-21T12:30:00Z"
}
,,,
```

Response '200 OK': profil KYC mis à jour.

'POST /customers/{id}/kyc/reassess'

Body: aucun.

Response '200 OK': objet 'Customer' après recalcul du risque.

```
'''json
{
  "id": 1,
  "full_name": "Grace Hopper",
  "id_number": "ID-123456",
  "id_type": "NATIONAL_ID",
  "phone": "+2438200000000",
  "address": "Gombe, Kinshasa",
  "created_at": "2026-08-21T12:30:00Z",
  "updated_at": "2026-08-21T12:35:00Z",
  "deleted_at": null,
  "risk_level": "LOW",
  "risk_score": 0
}
```

```
'''
```

```
### 'POST /transactions'
```

Body:

```
'''json
{
  "amount": 12500.75,
  "currency": "USD",
  "target_currency": "CDF",
  "type": "deposit",
  "rate": 2845.5,
  "status": "PENDING",
  "trader_id": 1,
  "customer_id": 1
}
'''
```

Response '201 Created': objet 'Transaction' apres conversion et analyse AML/KYC synchrone.

```
'''json
{
  "id": 10,
  "amount": 12500.75,
  "currency": "USD",
  "target_currency": "CDF",
  "type": "deposit",
  "rate": 2845.5,
  "converted_amount": 35571408.75,
  "risk_level": "MEDIUM",
  "risk_score": 30,
  "status": "PENDING",
  "created_at": "2026-08-21T12:30:00Z",
  "updated_at": "2026-08-21T12:30:01Z",
  "deleted_at": null,
  "trader_id": 1,
  "customer_id": 1
}
'''
```

```
### 'GET /transactions/{id}'
```

Body: aucun.

Response '200 OK': objet 'Transaction'.

```
### 'POST /risk-flags'
```

Body:

```
'''json
{
  "flag": "MANUAL_REVIEW",
  "reason": "Document KYC a verifier",
  "score": 15,
  "level": "MEDIUM",
  "transaction_id": 10,
}
```

```
    "trader_id": 1,
    "customer_id": 1
  }
  '''
```

Response '201 Created':

```
'''json
{
  "id": 3,
  "flag": "MANUAL_REVIEW",
  "reason": "Document KYC a verifier",
  "score": 15,
  "level": "MEDIUM",
  "transaction_id": 10,
  "trader_id": 1,
  "customer_id": 1,
  "created_at": "2026-08-21T12:30:00Z",
  "updated_at": "2026-08-21T12:30:00Z",
  "deleted_at": null
}
'''
```

'GET /risk-flags'

Body: aucun.

Query params optionnels:

- 'transaction_id'
- 'trader_id'
- 'customer_id'

Exemples:

```
'''http
GET /risk-flags
GET /risk-flags?transaction_id=10
GET /risk-flags?customer_id=1&trader_id=1
'''
```

Response '200 OK':

```
'''json
[
  {
    "id": 3,
    "flag": "MANUAL_REVIEW",
    "reason": "Document KYC a verifier",
    "score": 15,
    "level": "MEDIUM",
    "transaction_id": 10,
    "trader_id": 1,
    "customer_id": 1,
    "created_at": "2026-08-21T12:30:00Z",
    "updated_at": "2026-08-21T12:30:00Z",
    "deleted_at": null
  }
]
```

```
]
'''
```

```
### 'GET /risk-flags/{id}'
```

Body: aucun.

Response '200 OK': objet 'RiskFlag'.

```
### 'DELETE /risk-flags/{id}'
```

Body: aucun.

Response: '204 No Content'.

```
## Traders
```

```
### Creer Un Trader
```

```
'''http
POST /traders
Content-Type: application/json
'''
```

```
'''json
{
  "name": "Bureau Gombe",
  "first_name": "Jean",
  "last_name": "Kabila",
  "email": "jean@example.com",
  "phone": "+2438100000000",
  "status": "ACTIVE",
  "role": "TRADER",
  "password_hash": "$2a$...",
  "store_id": 1,
  "is_active": true
}
'''
```

Champs obligatoires:

- 'email'
- 'password_hash'

Reponse '201 Created':

```
'''json
{
  "id": 1,
  "name": "Bureau Gombe",
  "first_name": "Jean",
  "last_name": "Kabila",
  "email": "jean@example.com",
  "phone": "+2438100000000",
  "status": "ACTIVE",
  "created_at": "2026-08-21T12:30:00Z",
  "updated_at": "2026-08-21T12:30:00Z",
  "deleted_at": null,
}
```

```
    "role": "TRADER",
    "password_hash": "$2a$...",
    "store_id": 1,
    "is_active": true
}
```

Recuperer Un Trader

```
'''http
GET /traders/{id}
'''
```

Reponse '200 OK': objet 'Trader'.

Mettre A Jour Un Trader

```
'''http
PUT /traders/{id}
Content-Type: application/json
'''
```

Utilise le meme payload que la creation. Le handler ne modifie pas 'password_hash'.

Champs obligatoires:

- 'email'

Reponse '200 OK': objet 'Trader'.

Supprimer Un Trader

```
'''http
DELETE /traders/{id}
'''
```

Reponse '204 No Content'. La suppression est logique via 'deleted_at'.

Customers

Creer Un Client

```
'''http
POST /customers
Content-Type: application/json
'''
```

```
'''json
{
  "full_name": "Grace Hopper",
  "id_number": "ID-123456",
  "id_type": "PASSPORT",
  "phone": "+2438200000000",
  "address": "Kinshasa"
}
'''
```

Champs obligatoires:

- 'full_name'
- 'id_number'

Reponse '201 Created':

```
'''json
{
  "id": 1,
  "full_name": "Grace Hopper",
  "id_number": "ID-123456",
  "id_type": "PASSPORT",
  "phone": "+2438200000000",
  "address": "Kinshasa",
  "created_at": "2026-08-21T12:30:00Z",
  "updated_at": "2026-08-21T12:30:00Z",
  "risk_level": "",
  "risk_score": 0
}
'''
```

Creer Un Client Avec KYC

```
'''http
POST /customers/kyc
Content-Type: application/json
'''
```

```
'''json
{
  "customer": {
    "full_name": "Grace Hopper",
    "id_number": "ID-123456",
    "id_type": "PASSPORT",
    "phone": "+2438200000000",
    "address": "Kinshasa"
  },
  "kyc_profile": {
    "legal_nature": "INDIVIDUAL",
    "activity_profile": "BUSINESS",
    "profession": "Entrepreneur",
    "source_of_funds": "Business revenue",
    "purpose_of_operations": "Currency exchange",
    "expected_volume": 50000,
    "expected_frequency": 12
  }
}
'''
```

Notes:

- 'kyc_status' est force a 'UNVERIFIED' a la creation initiale.
- Un risque initial est calcule et persiste sur le client.
- Une personne morale 'LEGAL_ENTITY' ne peut pas avoir 'activity_profile = PERSONAL'.

Reponse '201 Created': objet 'Customer' avec 'risk_level' et 'risk_score'.

Recuperer Un Client

```
'''http
GET /customers/{id}
'''
```

Reponse '200 OK': objet 'Customer'.

Mettre A Jour Un Client

```
'''http
PUT /customers/{id}
Content-Type: application/json
'''
```

Utilise le meme payload que 'POST /customers'.

Reponse '200 OK': objet 'Customer'.

KYC

Recuperer Le Profil KYC

```
'''http
GET /customers/{id}/kyc
'''
```

Reponse '200 OK':

```
'''json
{
  "id": 1,
  "customer_id": 1,
  "legal_nature": "INDIVIDUAL",
  "activity_profile": "BUSINESS",
  "kyc_status": "VERIFIED",
  "profession": "Entrepreneur",
  "employer": "",
  "company_name": "",
  "registration_number": "",
  "source_of_funds": "Business revenue",
  "purpose_of_operations": "Currency exchange",
  "expected_volume": 50000,
  "expected_frequency": 12,
  "last_verification_date": "2026-08-21T12:30:00Z",
  "created_at": "2026-08-21T12:00:00Z",
  "updated_at": "2026-08-21T12:30:00Z"
}
'''
```

Mettre A Jour Le Profil KYC

```
'''http
PUT /customers/{id}/kyc
Content-Type: application/json
'''
```

```
'''json
{
```

```
"legal_nature": "INDIVIDUAL",
"activity_profile": "PROFESSIONAL",
"kyc_status": "VERIFIED",
"profession": "Consultant",
"employer": "Self-employed",
"source_of_funds": "Consulting income",
"purpose_of_operations": "FX operations",
"expected_volume": 25000,
"expected_frequency": 8,
"last_verification_date": "2026-08-21T12:30:00Z"
}
'''
```

Comportement:

- preserve 'id', 'customer_id' et 'created_at';
- conserve l'ancien 'kyc_status' si le champ est vide;
- conserve 'last_verification_date' si le champ est vide;
- valide la coherence KYC;
- persiste le profil;
- relance automatiquement 'TriggerRiskReassessment'.

Reponse '200 OK': profil KYC mis a jour.

Recalculer Le Risque KYC

```
'''http
POST /customers/{id}/kyc/reassess
'''
```

Reponse '200 OK': objet 'Customer' apres recalcul.

Enumerations KYC

'legal_nature':

- 'INDIVIDUAL'
- 'LEGAL_ENTITY'

'activity_profile':

- 'PERSONAL'
- 'PROFESSIONAL'
- 'BUSINESS'
- 'OTHER'

'kyc_status':

- 'UNVERIFIED'
- 'PARTIALLY_VERIFIED'
- 'VERIFIED'
- 'ENHANCED_VERIFICATION'

Transactions FX

Creer Une Transaction

```
'''http
```

```
POST /transactions
Content-Type: application/json
'''
```

```
'''json
{
  "amount": 12500.75,
  "currency": "USD",
  "target_currency": "CDF",
  "type": "deposit",
  "rate": 2845.5,
  "status": "PENDING",
  "trader_id": 1,
  "customer_id": 2
}
'''
```

Champ	Type	Obligatoire	Description
'amount'	number	oui	Montant source. Doit etre '> 0'.
'currency'	string	oui	Devise source, convertie en majuscules.
'target_currency'	string	oui	Devise cible, convertie en majuscules.
'type'	string	non	Type metier, par exemple 'deposit', 'withdrawal', 'buy', 'sell'.
'rate'	number	oui	Taux fixe par le trader. Doit etre '> 0'.
'status'	string	non	Statut initial.
'trader_id'	number	oui	Identifiant du trader.
'customer_id'	number	oui	Identifiant du client.

Calcul De Conversion

Le calcul est fait avant la persistance:

```
'''text
converted_amount = amount * rate
'''
```

Le resultat est arrondi a 4 decimales.

Exemple:

```
'''text
100 USD * 2850.55 = 285055 CDF
'''
```

Reponse '201 Created':

```
'''json
{
  "id": 10,
  "amount": 12500.75,
  "currency": "USD",
  "target_currency": "CDF",
  "type": "deposit",
  "rate": 2845.5,
  "converted_amount": 35571408.75,
  "status": "PENDING",
  "created_at": "2026-08-21T12:30:00Z",
  "updated_at": "2026-08-21T12:30:00Z",
}
```

```
"deleted_at": null,  
"trader_id": 1,  
"customer_id": 2  
}  
'''
```

Important: l'analyse AML/KYC est executee avant la reponse. La reponse '201 Created' contient donc le 'risk_level' et le 'risk_score' transactionnels finalises.

Recuperer Une Transaction

```
'''http  
GET /transactions/{id}  
'''
```

Reponse '200 OK': objet 'Transaction', avec 'risk_level' et 'risk_score'.

Risk Flags

Les 'risk_flags' representent les alertes produites par le moteur AML/KYC ou ajoutees manuellement.

Creer Un Risk Flag Manuel

```
'''http  
POST /risk-flags  
Content-Type: application/json  
'''
```

```
'''json  
{  
  "flag": "MANUAL_REVIEW",  
  "reason": "Document KYC a verifier",  
  "score": 15,  
  "level": "MEDIUM",  
  "transaction_id": 10,  
  "trader_id": 1,  
  "customer_id": 2  
}  
'''
```

Champs obligatoires:

- 'flag'
- 'transaction_id'

Reponse '201 Created':

```
'''json  
{  
  "id": 3,  
  "flag": "MANUAL_REVIEW",  
  "reason": "Document KYC a verifier",  
  "score": 15,  
  "level": "MEDIUM",  
  "transaction_id": 10,  
  "trader_id": 1,  
  "customer_id": 2,  
}
```

```
"created_at": "2026-08-21T12:30:00Z",
"updated_at": "2026-08-21T12:30:00Z",
"deleted_at": null
}
'''
```

Lister Les Risk Flags

```
'''http
GET /risk-flags
GET /risk-flags?transaction_id=10
GET /risk-flags?customer_id=2
GET /risk-flags?trader_id=1
'''
```

Filtres optionnels combinables:

- 'transaction_id'
- 'customer_id'
- 'trader_id'

Reponse '200 OK': tableau de 'RiskFlag', trie par 'created_at DESC, id DESC'.

Recuperer Un Risk Flag

```
'''http
GET /risk-flags/{id}
'''
```

Reponse '200 OK': objet 'RiskFlag'.

Supprimer Un Risk Flag

```
'''http
DELETE /risk-flags/{id}
'''
```

Reponse '204 No Content'. La suppression est logique via 'deleted_at'.

Analyse AML/KYC

'RiskService.AnalyzeTransactionAndCustomer' est execute pendant chaque 'POST /transactions', apres la persistance de la transaction et avant la reponse HTTP.

Flux:

1. recuperation du client;
2. recuperation du profil KYC si disponible;
3. recuperation de l'historique transactionnel du client sur 30 jours;
4. execution des regles transactionnelles;
5. mise a jour du risque transaction;
6. execution des regles KYC/client;
7. creation des 'risk_flags' pour toutes les regles declenchees;
8. mise a jour du risque global client avec 'transactionRisk.Score + customerRisk.Score'.

Regles Transactionnelles

| Code | Condition | Score |

---	---	---
'HIGH_TRANSACTION_AMOUNT'	'amount > 10000.0'	'30'
'CUMULATIVE_VOLUME_24H'	cumul client sur 24h '> 20000.0'	'35'
'CUMULATIVE_VOLUME_30D'	cumul client sur 30j '> 100000.0'	'40'
'UNUSUAL_FREQUENCY_24H'	nombre de transactions client sur 24h '> 10'	'25'

Seuils du risque transaction:

Score	Niveau
---	---
'>= 75'	'CRITICAL'
'>= 50'	'HIGH'
'>= 25'	'MEDIUM'
'< 25'	'LOW'

Regles KYC/Client

Code	Condition	Score
---	---	---
'INSUFFICIENT_KYC_STATUS'	profil KYC absent ou statut different de 'VERIFIED' / 'ENHANCED_VERIFICATION'	'45'
'INCONSISTENT_KYC_PROFILE'	profil absent, personne morale 'PERSONAL', donnees societe manquantes, profession manquante ou source/purpose manquants	'35'
'BUSINESS_ACTIVITY_PROFILE'	'activity_profile == BUSINESS'	'15'
'EXPECTED_VOLUME_EXCEEDED'	cumul client 30j '> expected_volume' declare au KYC	'30'

Seuils du risque KYC/client:

Score	Niveau
---	---
'>= 70'	'CRITICAL'
'>= 40'	'HIGH'
'>= 20'	'MEDIUM'
'< 20'	'LOW'

Seuils du risque global client apres transaction:

Score total	Niveau
---	---
'>= 90'	'CRITICAL'
'>= 60'	'HIGH'
'>= 30'	'MEDIUM'
'< 30'	'LOW'

Recalcul KYC Manuel

'POST /customers/{id}/kyc/reassess' utilise le profil KYC courant du client et met a jour 'customers.risk_score' / 'customers.risk_level'.

Ce recalcul manuel est plus simple que l'analyse AML post-transaction: il ne lance pas les regles de cumul/frequence transactionnelle.

Modele De Donnees

'traders'

Colonne	Type	Notes
---	---	---

	'id'		'SERIAL'		Cle primaire.	
	'name'		'VARCHAR(255)'		Nom affiche.	
	'first_name'		'VARCHAR(255)'		Prenom.	
	'last_name'		'VARCHAR(255)'		Nom.	
	'email'		'VARCHAR(255)'		Unique, obligatoire.	
	'phone'		'VARCHAR(50)'		Telephone.	
	'status'		'VARCHAR(50)'		Statut.	
	'role'		'VARCHAR(50)'		Role.	
	'password_hash'		'VARCHAR(255)'		Obligatoire.	
	'store_id'		'INT'		Point de vente.	
	'is_active'		'BOOLEAN'		Default 'TRUE'.	
	'created_at', 'updated_at', 'deleted_at'		'TIMESTAMPTZ'		Timestamps.	

'customers'

	Colonne		Type		Notes	
	---		---		---	
	'id'		'SERIAL'		Cle primaire.	
	'full_name'		'VARCHAR(255)'		Obligatoire.	
	'id_number'		'VARCHAR(100)'		Unique, obligatoire.	
	'id_type'		'VARCHAR(50)'		Type de piece.	
	'phone'		'VARCHAR(50)'		Telephone.	
	'address'		'TEXT'		Adresse.	
	'risk_level'		'VARCHAR(50)'		'LOW', 'MEDIUM', 'HIGH', 'CRITICAL'.	
	'risk_score'		'NUMERIC(5, 2)'		Default '0'.	
	'created_at', 'updated_at', 'deleted_at'		'TIMESTAMPTZ'		Timestamps.	

'customer_kyc_profiles'

	Colonne		Type		Notes	
	---		---		---	
	'id'		'SERIAL'		Cle primaire.	
	'customer_id'		'INT'		Unique, reference 'customers(id)'.	
	'legal_nature'		'VARCHAR(50)'		'INDIVIDUAL' ou 'LEGAL_ENTITY'.	
	'activity_profile'		'VARCHAR(50)'		'PERSONAL', 'PROFESSIONAL', 'BUSINESS', 'OTHER'.	
	'kyc_status'		'VARCHAR(50)'		Statut KYC.	
	'profession', 'employer'		'VARCHAR(255)'		Champs personne physique.	
	'company_name', 'registration_number'		'VARCHAR(255)'		Champs personne morale.	
	'source_of_funds'		'TEXT'		Source des fonds.	
	'purpose_of_operations'		'TEXT'		Objectif des operations.	
	'expected_volume'		'NUMERIC(14, 2)'		Volume mensuel attendu.	
	'expected_frequency'		'INT'		Nombre de transactions mensuelles attendu.	
	'last_verification_date'		'TIMESTAMPTZ'		Derniere verification.	
	'created_at', 'updated_at'		'TIMESTAMPTZ'		Timestamps.	

'transactions'

	Colonne		Type		Notes	
	---		---		---	
	'id'		'SERIAL'		Cle primaire.	
	'amount'		'NUMERIC(19, 4)'		Montant source.	
	'currency'		'VARCHAR(10)'		Devise source.	
	'target_currency'		'VARCHAR(10)'		Devise cible.	
	'type'		'VARCHAR(50)'		Type de transaction.	
	'rate'		'NUMERIC(19, 6)'		Taux fixe par le trader.	
	'converted_amount'		'NUMERIC(19, 4)'		Resultat 'amount * rate'.	
	'risk_level'		'VARCHAR(50)'		Mis a jour par l'analyse AML.	
	'risk_score'		'NUMERIC(5, 2)'		Mis a jour par l'analyse AML.	

```
| 'status' | 'VARCHAR(50)' | Statut transactionnel. |
| 'trader_id' | 'INT' | Reference 'traders(id)'. |
| 'customer_id' | 'INT' | Reference 'customers(id)'. |
| 'created_at', 'updated_at', 'deleted_at' | 'TIMESTAMPTZ' | Timestamps. |
```

'risk_flags'

```
| Colonne | Type | Notes |
| --- | --- | --- |
| 'id' | 'SERIAL' | Cle primaire. |
| 'flag' | 'VARCHAR(255)' | Code de l'alerte. |
| 'reason' | 'TEXT' | Explication lisible. |
| 'score' | 'NUMERIC(5, 2)' | Score associe. |
| 'level' | 'VARCHAR(50)' | Niveau optionnel. |
| 'transaction_id' | 'INT' | Transaction concerne. |
| 'trader_id' | 'INT' | Trader concerne. |
| 'customer_id' | 'INT' | Client concerne. |
| 'created_at', 'updated_at', 'deleted_at' | 'TIMESTAMPTZ' | Timestamps. |
```

Tests

```
'''bash
go test ./...
'''
```

Dans un environnement ou le cache Go utilisateur n'est pas accessible:

```
'''bash
GOCACHE=/private/tmp/fx-app-api-go-build go test ./...
'''
```

Limitations Actuelles

- 'POST /transactions' persiste la transaction avant l'analyse AML/KYC; une erreur d'analyse apres insertion n'est pas encore rollbackee dans une transaction SQL.
- Les regles AML sont codees en dur, pas encore configurees en base ou fichier.
- 'RuleConfig.Enabled' existe mais n'est pas encore utilise pour desactiver une regle.
- Pas encore de screening sanctions, PEP, listes noires ou adverse media.
- Pas encore de workflow d'investigation complet pour les alertes ('OPEN', 'IN_REVIEW', 'RESOLVED', 'FALSE_POSITIVE').
- Pas encore de gestion documentaire KYC: documents, expiration, verification de piece, justificatif d'adresse.
- Pas encore d'authentification/RBAC actif sur les routes.
- Les erreurs de contraintes SQL sont encore renvoyees en '500'.
- Les formats metier de 'currency', 'status', 'type', 'role' ne sont pas encore normalises par enum stricte.